NNN-E-00920b
16 February 1967

Interim Revision of
Fed. Spec. NNN-E-920a
19 February 1963

INTERIM FEDERAL SPECIFICATION

EXTRACTION ASSEMBLY

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification NNN-E-920a.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a Soxhlet extractor, a boiling flask, and a Friedrichs condenser suitable for constructing an extraction assembly for laboratory use.

1.2 Classification. The extractor, flask, and condenser shall be of the following types, sizes, and styles, as specified (see 6.2).

Type I - Soxhlet extraction tube

Size 1. 50 ml. capacity siphon tube
Size 2. 85 ml. capacity siphon tube
Size 3. 200 ml. capacity siphon tube

Type II - Boiling flask

Style A. Round bottom
Style B. Flat bottom

Size 1. 100 ml. capacity
Size 2. 125 ml. capacity
Size 3. 250 ml. capacity
Size 4. 500 ml. capacity

Type III - Friedrichs condenser

Style A. Side arm, accommodates No. 3 Rubber Stopper
Style B. Side arm, 24/40 Standard Taper Joint

Size 1. Bottom inner joint size 24/40
Size 2. Bottom inner joint size 34/45

FSC 6640

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2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS.

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

DD-G-541	Glass (Laboratory).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver San Franscisco, Los Angeles, Seattle,and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Commercial Standard CS-21 Interchangeable Taper-Ground Joints, Stopcocks, Stoppers, and Spherical-Ground Joints

(Copies may be obtained from Superintendent of Documents, U.S. Government Printing Office, Washington 25, D. C.)

American Society for Testing and Materials Standards

ASTM Designation C-148 Polariscopic Examination of Glass Containers

(Copies of the ASTM may be obtained from The American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 Material. The Soxhlet extractor, the boiling flask and the Friedrichs condenser shall be fabricated of borosilicate glass conforming to type I, class A of DD-G-541.

3.1.1 Ground glass joint. The standard taper joints shall conform with dimensional requirements of Commercial Standard CS-21.

3.2 Soxhlet extractor.

3.2.1 Style and design. The Soxhlet extractor shall conform to figure 1 for style and design.

3.2.2 Dimensions. The Soxhlet extractor shall conform to the dimensional requirements specified in Table I.

Table I. Dimensions of extractor

	Inside diameter of tube, mm. ± 3 mm.	Total length of body, mm. ± 8 mm.	Listed capacity to top of siphon, ml.	Standard taper joint	
				Top	Bottom
Small	30	185	50	34/45	24/40
Medium	40	205	85	45/50	24/40
Large	50	243	200	55/50	24/40

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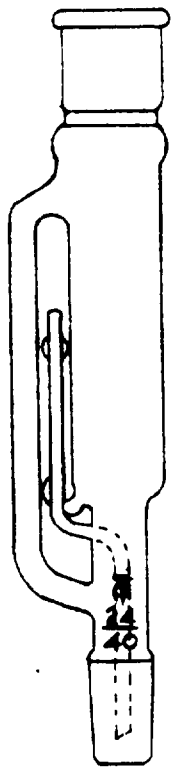


FIGURE 1 SOXHLET EXTRACTOR

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3.3 Boiling flask.

3.3.1 Style and design. The boiling flask shall be the short-neck type, having a flat bottom or round bottom, as specified (see 6.2). An area for marking shall be provided by blasting or spot enamel. The enameled area shall show no change or removal of enamel and shall be satisfactory for marking after testing as specified in 4.4.1. The boiling flasks shall conform to figure 2 for style and design.

3.3.2 Dimensions. The boiling flask shall conform to the dimensional requirements specified in Table II.

Table II. Dimensions for flasks

Size	Capacity (Milliliters)	Overall Height Style A	Overall Height Style B	Standard Taper Joint
1	100 ml.	120 mm.	--	24/40
2	125 ml.	--	115 mm.	24/40
3	250 ml.	140 mm.	130 mm.	24/40
4	500 ml.	160 mm.	155 mm.	24/40

3.4 Friedrichs condenser.

3.4.1 Style and design. The Friedrichs condenser shall be provided with a side arm that will accommodate a number 3 rubber stopper, (style A), or a side arm that has a 24/40 standard-taper inner joint (style B), as specified (see 6.2). The condenser shall conform to figure 3 for style and design.

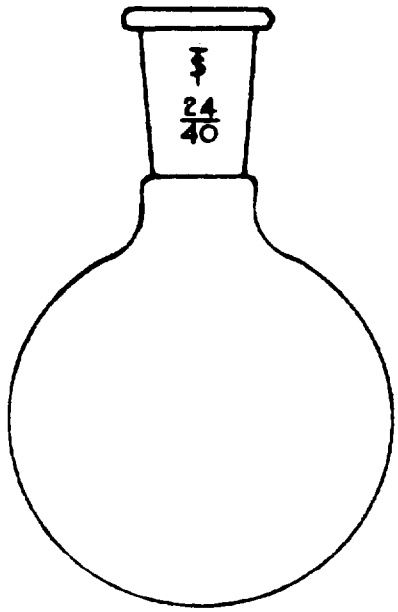
3.4.2 Dimensions. The condenser shall conform with the dimensional requirements as specified in figure 3.

3.5 Annealing. The ring strain for all components of the extraction assembly shall be not greater than temper No. 1 when tested as specified in 4.4.2. Longitudinal strain if present shall be faint and highly diffused.

3.6 Identification marking. All components of the extraction assembly shall be permanently and legibly marked with the manufacturer's name or registered trademark. In addition, all standard-taper ground joints shall be permanently and legibly marked with the proper joint size.

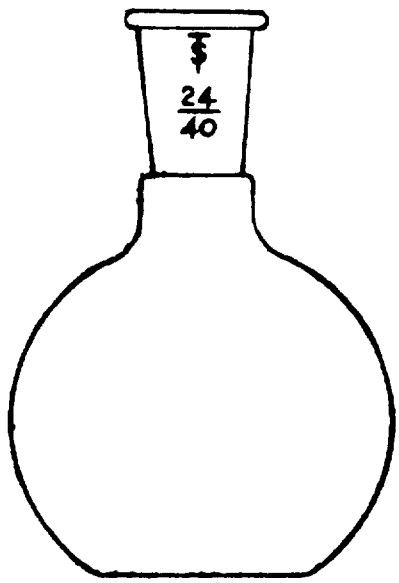
3.7 Workmanship. All components of the extraction assembly shall be properly annealed and finished. All components shall be free from defects which detract from their appearance or may impair their serviceability.

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ROUND BOTTOM

STYLE A

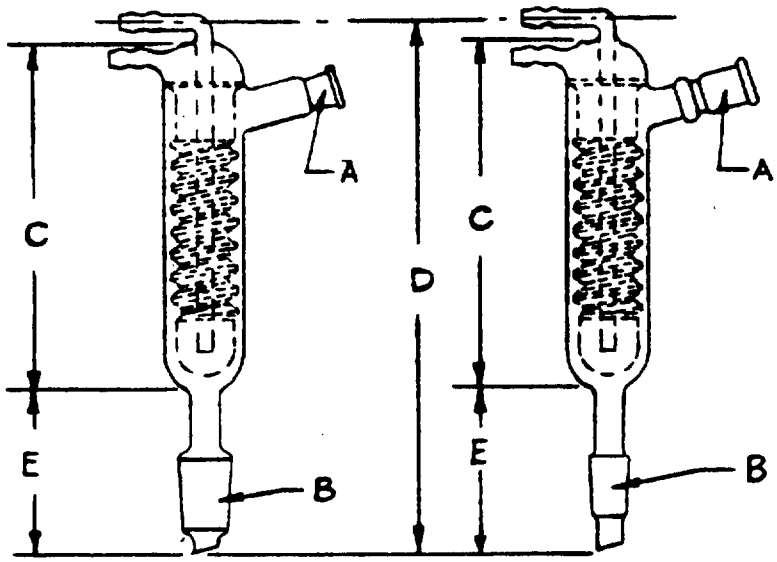


FLAT BOTTOM

STYLE B

FIGURE 2 BOILING FLASK

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STYLE A STYLE B

STYLE	A FEMALE JOINT SIZE	B MALE JOINT SIZE	C JACKET LENGTH MM.	D OVERALL LENGTH MM.	E STEM LENGTH MM.
A	NO. 3 RUBBER STOPPER	34/45	225	$340^{\pm 15}$	$90^{\pm 15}$
B	24/40	24/40	225	$335^{\pm 20}$	$90^{\pm 20}$

FIGURE 3 FRIEDRICHS CONDENSER

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier shall utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Total (Major and minor combined defects)	II	4.0
For dimensional examination	S-1	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 for inspection level S-1. Acceptance number shall be zero.

4.3 Examination. The Friedrichs condenser, the Soxhlet extractor and the boiling flasks shall be examined to determine compliance with all requirements contained in this specification.

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4.3.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

Categories and defects *

Major

101 Components not free from cracks, chips and sharp edges.
102 Components not properly annealed.
103 Components not free from bubbles or gaseous inclusions.
104 Components not symmetrically formed with the vertical axis.
105 Exposed edges of all components not fire polished.

Minor

201 Identification marking not complete, permanent or legible.
202 Components not free from dirt, grease, or other foreign particles.

* Examination shall not be restricted to the classified possible defects listed above.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Area of marking permanency.

4.4.1.1 Alkali test. Immerse in normal sodium hydroxide at room temperature, cover, and boil for 1/2 hour, and let stand for 1 hour. Remove flask and rinse thoroughly with distilled water. Dry by rubbing vigorously with a cloth.

4.4.1.2 Acid test (sulfuric). Immerse in concentrated sulfuric acid (sp gr 1.84) at room temperature. Let stand 1 hour. Remove flask and rinse thoroughly with distilled water. Dry by rubbing vigorously with a cloth.

4.4.2 Annealing. All components shall be tested for ring strain in accordance with ASTM Designation C-148.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. Each condenser, each extractor, and each flask of the type and size specified, or each extraction assembly, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each condenser, each extractor, and each flask shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure of carton shall be adequate to prevent spilling of contents under normal handling.

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5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. One extraction assembly or four condensers, extractors, or flasks of one type and size shall be packed in an exterior container designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. One extractor assembly or four condensers, extractors, or flasks of one type and size, shall be packed in an exterior container conforming to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, marking shall be in accordance with FED-STD-123.

5.3.2 Military agencies.

5.3.2.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers a Soxhlet extractor, a boiling flask, and a Friedrichs condenser suitable for constructing an extraction assembly intended to be used for the laboratory determination of the constituents in such substances as fats, oils, and gums which are soluble in volatile liquids.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Types, sizes, and style of extractor, flask and condenser required.

(c) Round bottom or flat bottom flask as required.

(d) The style of side arm required for the Friedrichs condenser.

(e) Selection of applicable levels of packing (see 5.2.1 and 5.2.2).

(f) Selection of applicable marking requirements (see 5.3.1 and 5.3.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MD
Air Force - 03

Code N

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Defense Supply Agency - DM

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